

# SUSHIL KUMAR AMMANAGHATTA SHIVAKUMAR

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## Education

### Albert Ludwig University of Freiburg

*Master of Science in Computer Science (Specialization in AI)*

- Thesis: *A Generative Model for Video Montage Creation*

Oct. 2021 – Oct. 2024

Freiburg, Germany

### PESIT-BSC

*Bachelor of Engineering in Computer Science*

Aug. 2017 – June 2021

Bengaluru, India

## Research Interests

*Biometric security and privacy, with a focus on backdoor attacks in biometric systems.*

## Work/Research Experience

### University of Paderborn

*Research Associate (Advisor: Prof. Dr. Philipp Terhorst)*

- Developing and analyzing backdoor attacks and corresponding defenses for open-set face recognition systems
- Tutored exercise classes for the course "Machine Learning for Biometrics"

Nov 2025 – May 2026

Paderborn, Germany

### Max Planck Institute for Intelligent Systems

*Research Intern (Advisor: Prof. Dr. Antonio Orvieto)*

- Long-Range Music Generation with focus on coherence and structure in symbolic compositions.
- Applied REMI+ event-based representation and BPE compression using miditok to preprocess MIDI sequences for long-range generative modeling.
- Trained MAMBA-based architectures for autoregressive music generation with large token windows.

Nov 2024 – April 2025

Tübingen, Germany

### Uni Freiburg, CV Lab(with Zebracat AI)

*Master Thesis Student (Advisor: Mohammadreza Zolfaghar)*

- Developed generative framework for video montage creation by aligning video and text embeddings.
- Used GPT and UMT to reframe classification into regression-based embedding alignment.
- Achieved new benchmarks on VSPD dataset (IoU: 0.167, UMS: 1.257, SMS: 0.103).
- Validated montage-aligned video generation quality through qualitative retrieval analysis.

Feb 2024 – Oct 2024

Freiburg, Germany

### Fraunhofer Institute for Solar Energy Systems ISE

*Research Assistant (Advisor: Dr. Paul Gebhard)*

- Automated scraping pipelines for scientific literature using Selenium and PyMuPDF.
- Developed parsers to extract structured metadata (tables, figures, references) from PDFs.
- Built RAG pipelines using Azure OpenAI for semantic data extraction on PV degradation.
- Implemented FAISS-based document indexing for efficient semantic retrieval.

May 2023 – Oct 2024

Freiburg, Germany

### Max Planck Institute for Security and Privacy (MPI-SP)

*Research Assistant (Advisor: Prof. Dr. Asia J. Biega)*

- Conducted empirical audit on deceptive GDPR patterns in top 10k Tranco-ranked websites.
- Built automation scripts to capture consent banners and identify 'Legitimate Interest' cases.
- Performed qualitative annotation using MAXQDA across screenshots and public forum discussions.

Apr 2022 – Mar 2023

Bochum, Germany

## Projects

### Training Noisy Real vs. Generated Images for Attribute Classification | Nov 2023

- Evaluated OpenCLIP performance on real-world noisy images compared to synthetic counterparts generated using Stable Diffusion.
- Created attribute-specific datasets focused on material, pattern, group, and color classification using CLIP-retrieved samples from the LAION dataset and synthetic images generated via Stable Diffusion.
- Found real noisy images offered better generalization due to inherent visual complexity.

## Publications

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### Investigating Deceptive Design in GDPR's Legitimate Interest

*ACM CHI 2023*

- Authors: Lin Kyi, **Sushil Ammanaghatta Shivakumar**, Franziska Roesner, Cristiana Santos, Frederike Zufall, and Asia Biega.
- Stefano Rodotà Award

### Turning to Online Forums for Legal Information

*Annual Privacy Forum 2025*

- Authors: Lin Kyi, Cristiana Santos, **Sushil Ammanaghatta Shivakumar**, Franziska Roesner, and Asia Biega.
- Full title: *Turning to Online Forums for Legal Information: A Case Study of GDPR's Legitimate Interests.*

### AutoML Decathlon: Diverse Tasks, Modern Methods, and Efficiency at Scale

*NeurIPS 2022 CT*

- **Competition Track**

## Technical Skills

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**Languages:** Python, PyTorch, LaTeX

**Developer Tools:** Git, SLURM, HTCondor

**Technologies / Libraries:** FAISS, PyMuPDF, Selenium, MAXQDA

**Relevant Coursework:** Machine Learning, Deep Learning, Information Retrieval, Mobile Robotics, Robot Mapping